Cognitive Load Modulates the Neural Mechanisms of Syntax–Plausibility Integration: An fMRI Study

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Introduction

This study investigates how sentence processing dynamically adapts under working memory constraints. Humans derive sentence meaning by integrating multiple sources of information, including grammar and semantic plausibility (Gibson & Pearlmutter, 1998). This integration becomes more challenging when working memory resources are limited (Ayasse et al., 2021; Gibson et al., 2015).

However, there is little evidence regarding whether the integration of syntactic and plausibility information is modulated "during sentence processing" rather than during post-sentence comprehension. Behavioral findings from prior work may reflect processes at the question-answering stage rather than online comprehension.

The present study examines how cognitive load modulates syntax–plausibility integration using functional Magnetic Resonance Imaging (fMRI). Participants performed a dual-task paradigm while listening to plausible and implausible sentences.

In summary, the findings highlight a key role for the Left Inferior Frontal Gyrus (LIFG):

1. Under limited cognitive resources, LIFG pars triangularis (LIFGtri) adjusts the memory constraints on integration of syntactic and plausibility information.
2. People with higher working memory capacity (WMC) did not require increased activation in LIFGtri when processing implausible sentences during the dual-task condition.

Methods

We recruited 62 native Japanese speakers and excluded four participants due to excessive head motion (> 5 mm or > 10°) or low task accuracy (< 80%).

We conducted a dual-task paradigm, inspired by Rogalsky et al. (2013). As shown in Figure 1, participants either silently repeated meaningless syllables "da ba ga da…" (**Articulation**) or remained silent (**NoTask**) while listening to sentences. We additionally manipulated plausibility such as (1-2). After each sentence, participants answered a two-choice comprehension question like (3). Participants listened to 48 target stimuli and 48 filler sentences (all plausible) per person. After scanning, we assessed their WMC using the Operation Span Task (von der Malburg, 2015).

(1) Plausible: *赤ちゃんが母親に優しく抱き上げられた。*"A baby was gently lifted by a mother."

(2) Implausible: *母親が赤ちゃんに優しく抱き上げられた。* "A mother was gently lifted by a baby."

(3) Question: *誰が抱き上げたか？* "Who lifted?" → *赤ちゃん / 母親* "Baby / Mother"

Results and Discussion

Figure 2 shows whole-brain contrasts between plausible and implausible sentences. LIFGtri, LIFG pars orbitalis (LIFGorb), and Superior Frontal Gyrus showed significantly greater activation for implausible than plausible sentences across task conditions (cluster-level FWE-corrected $p < .05$). This aligns with previous works (Sassa et al., 2007; Ye & Zhou, 2009).

As shown in Figure 3, functional ROI analysis revealed that LIFGtri was significantly more activated in the dual-task than the single-task condition (FWE-corrected $p < .05$), whereas LIFGorb was not. These results suggest that LIFGtri plays a critical role in integrating syntactic and plausibility information when cognitive resources are restricted.

Finally, we analyzed individual differences by covariate of WMC score with activation of LIFGtri and LIFGorb of each region. Figure 4 indicates negative correlation between WMC and LIFGtri activation in the Implausible–Articulation condition ($r = -.40$, $p = .002$). This indicates that participants with higher WMC require less neural recruitment to process implausible sentences even under the cognitive load.

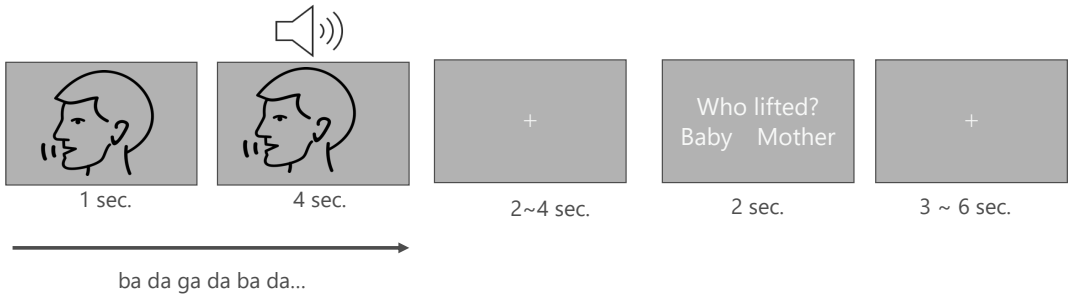
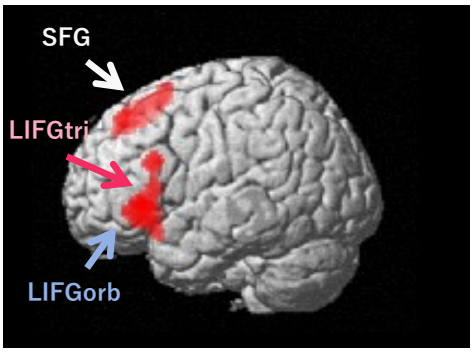


Figure 1. Procedure of each trial.



$p < 0.001$, cluster-size FWE $p < 0.05$

Figure 2. Main effect of plausibility Implausible > Plausible.

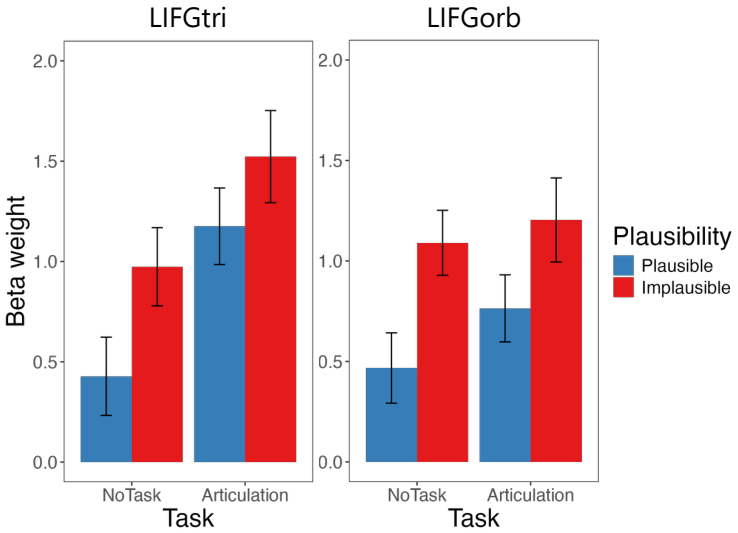


Figure 3. Beta weight of LIFGtri and LIFGorb in each condition.

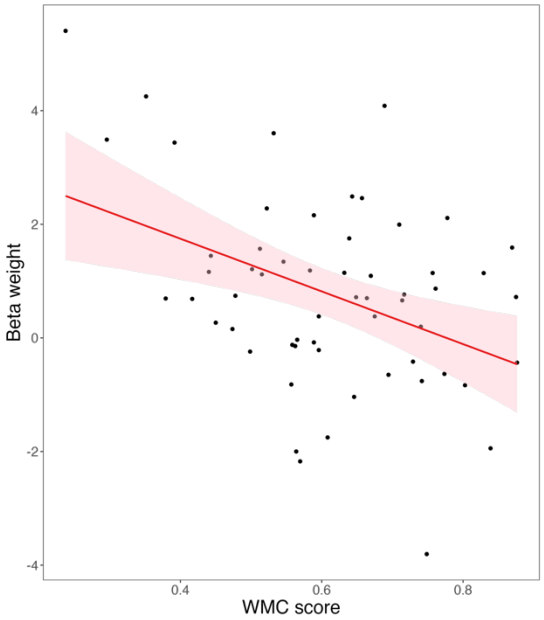


Figure 4. Covariate of working memory capacity (WMC) score with brain activation in the Articulation-Implausible condition.

References

Ayasse et al. (2021). The Principle of Least Effort and Comprehension of Spoken Sentences by Younger and Older Adults. *Frontiers in Psychology*.

Gibson & Pearlmutter. (1998). Constraints on sentence comprehension. *Trends in Cognitive Sciences*.

Gibson et al. (2015). A rational inference approach to aphasic language comprehension. *Aphasiology*.

Rogalsky et al. (2008). Broca's area, sentence comprehension, and working memory: an fMRI study *Frontiers in Human Neuroscience*.

Sassa et al. (2007). Processing of Anomalous Sentences in Japanese: An fMRI Study. *Journal of Cognitive Science*.

Ye & Zhou. (2009). Conflict control during sentence comprehension: fMRI evidence. *Neuroimage*.

Von der Malsburg. (2015). Py-Span-Task -- A Software for Testing Working Memory Span.